UNIVERSITY SEMESTER EXAMINATION — TOC

Subject: Theory of Computation (TOC) | Maximum Marks: 100

Section A & B Questions

Q1. (a) Define DFA formally as a 5-tuple. What is a transition function? (5 Marks)

Q1. (b) Differentiate between DPDA and NPDA. (5 Marks)

Q2. (a) State and prove Pumping Lemma for Regular Languages with L={0^n 1^n}. (10 Marks)

Q2. (b) Design a Turing Machine to accept palindromes over {a, b}. (10 Marks)